

Chapter

Revision

**1. In the statement $t = t + (i + x);$, which variable acts as the accumulator?
SQP 2021**

- (a) t
- (b) i
- (c) x
- (d) None

**2. How many bytes of memory does the int data type occupy in Java ?
MAIN 2023**

- (a) 4 bytes
- (b) 8 bytes
- (c) 2 bytes
- (d) 1 byte

**3. What is the total number of primitive data types available in Java?
SQP 2022**

- (a) 4
- (b) 6
- (c) 8
- (d) 2

4. Which operator has higher precedence in an expression containing both && and || in Java? COMP 2023

- (a) &&
- (b) ||
- (c) Both have equal precedence
- (d) None

5. Which of the following is the correct way to use the Math class to find the square root of a number? MAIN 2021

- A. Double result = Math.sqrt(16);
- B. Double result = Math.square(16);
- C. Double result = sqrt.Math(16);
- D. Double result = square.Math(16);

- (a) Only A
- (b) Only B
- (c) Both A and C
- (d) All the three

6. Assertion (A) : The switch construct in Java is also known as a fall-through construct.

Reason (R) : The break statement is mandatory after every case in a switch construct.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true. **SQP 2023**

7. Assertion (A) : The keyword new in Java is used to allocate memory for an object at runtime.

Reason (R) : The new keyword returns the reference (address) of the allocated memory.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true. **SQP 2021**

8. Assertion (A) : The Scanner class is used to take user input in Java.

Reason (R) : The Scanner class supports multiple data types, such as int, double, and char.

- (a) Both A and R are true, and R is the correct explanation of A.
 - (b) Both A and R are true, but R is not the correct explanation of A.
 - (c) A is true, but R is false.
 - (d) A is false, but R is true.
- COMP 2023**

9. What is object-oriented programming? SQP 2023

10. Name two Process Oriented languages and two Object Oriented languages.
MAIN 2022

11. Name any two principles/characteristics of Object Oriented Programs.
MAIN 2023

12. Write the usage of different OPs concepts. COMP 2021

13. Assertion (A) : Java is a platform-independent programming language.
Reason (R) : Java programs are compiled into byte code, which is executed by the Java Virtual Machine (JVM). **COMP 2022**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

14. Assertion (A) : Object-Oriented Programming (OP) is focused on procedures and logic.

Reason (R) : OP uses concepts like encapsulation, inheritance, and to model real-world entities. **MAIN 2021**

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

15. Identify the literals listed below: COMP 2023

- (i) 0.5
- (ii) 'A'
- (iii) false
- (iv) "a"

Solution

1. (a) It reacts with ethanol to form a sweet smelling compound

Ethanoic acid (CH_3COOH) reacts with ethanol ($\text{C}_2\text{H}_5\text{OH}$) in the presence of concentrated sulphuric acid to form ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$), which is a sweet-smelling ester, along with water.

2. (a) 4 bytes

In Java, the `int` data type always occupies 4 bytes (32 bits) of memory. It stores integer values in the range -2^{31} to $+2^{31} - 1$.

3. (c) 8

Explanation:

Java has 8 primitive data types. These are the basic data types used to store simple values.

4. (a) `&&`

Explanation:

In Java, the logical AND (`&&`) operator has higher precedence than the logical OR (`||`) operator.

- `&&` is evaluated before `||` in expressions without parentheses.

5. (a) Only A

The correct method to find the square root in Java using the `Math` class is `Math.sqrt(16)`; Options B, C, and D are invalid methods.

6. (c) A is true, but R is false

The switch construct is called a fall-through construct because without `break`, control moves to the next case. However, `break` is not mandatory, making R false.

7. (a) Both A and R are true, and R is the correct explanation of A

The new keyword dynamically allocates memory and returns its reference, making R the correct explanation for A.

8. (a) Both A and R are true, and R is the correct explanation of A.

The Scanner class allows for reading input in various data types, making R the correct explanation of A.

9. Object-oriented programming (OOP) is a programming paradigm that **simulates the real world in software**, where the real world is represented using **objects**.

10. Two Process-Oriented Languages: C, Fortran

Two Object-Oriented Languages: C++, Java

11. 1. Encapsulation – Bundling of data and methods that operate on the data into a single unit (class).

2. Inheritance – Ability of a class to acquire properties and behaviors of another class.

12. Usage of different OOP concepts:

- **Encapsulation:** Leads to abstraction and **protects the object from misuse** by restricting direct access to some of its components.
- **Abstraction:** Hides complex implementation details and **prevents mishandling or misuse** of the object.
- **Inheritance:** Allows **derived classes to inherit features** of the base class, avoiding the need for redefinition and promoting code reuse.
- **Polymorphism:** Enables **method/function overloading or overriding**, allowing the same method name to behave differently in different contexts.

13. (a) Both A and R are true, and R is the correct explanation of A

- **Assertion (A):** True — Java is **platform-independent** because the same Java program can run on any operating system.

- **Reason (R):** True — Java programs are compiled into **byte code**, which is executed by the **Java Virtual Machine (JVM)** on any platform.

14. (d) A is false, but R is true

- **Assertion (A):** False — Object-Oriented Programming (OOP) **focuses on objects**, not just procedures and logic.
- **Reason (R):** True — OOP uses concepts like **encapsulation, inheritance, and polymorphism** to model real-world entities.

15. (i) 0.5 – Floating literal

(ii) 'A' – Character literal

(iii) false – Boolean literal

(iv) "a" – String literal